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Ex: 1

$$A = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 & i \\ -i & 2 \end{pmatrix} \quad |\psi\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \Delta A \Delta B \geq \frac{1}{2} |\langle [A, B] \rangle|$$

$$\begin{aligned} [A, B] &= AB - BA = \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} \begin{pmatrix} 0 & i \\ -i & 2 \end{pmatrix} - \begin{pmatrix} 0 & i \\ -i & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 0 \end{pmatrix} \\ &= \begin{pmatrix} -2i & i+i \\ 0 & 2i \end{pmatrix} - \begin{pmatrix} 2i & 0 \\ -i+i & -2i \end{pmatrix} = \begin{pmatrix} -4i & i+i \\ i-i & 4i \end{pmatrix} \end{aligned}$$

$$\langle \psi | [A, B] | \psi \rangle = (1 \ 0) \begin{pmatrix} -4i & i+i \\ i-i & 4i \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = (1 \ 0) \begin{pmatrix} -4i \\ i-i \end{pmatrix} = -4i$$

$$\Delta A \Delta B \geq \frac{1}{2} |-4i| = \frac{1}{2} \cdot 4 = 2$$

$$\Delta A \Delta B \geq 2$$

Ex: 2

$$|\psi\rangle = \sqrt{\frac{2}{3}} |z_+\rangle + \frac{1}{\sqrt{3}} |z_-\rangle \quad \text{on } \{|z_+\rangle, |z_-\rangle\} \quad \begin{cases} |z_+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}} |z_+\rangle + \frac{1}{\sqrt{2}} |z_-\rangle \\ |z_-\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{\sqrt{2}} |z_+\rangle - \frac{1}{\sqrt{2}} |z_-\rangle \end{cases}$$

$$\begin{aligned} |\psi\rangle &= \sqrt{\frac{2}{3}} \cdot \frac{1}{\sqrt{2}} |z_+\rangle + \sqrt{\frac{2}{3}} \cdot \frac{1}{\sqrt{2}} |z_-\rangle + \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} |z_+\rangle - \frac{1}{\sqrt{3}} \cdot \frac{1}{\sqrt{2}} |z_-\rangle = \frac{1}{\sqrt{3}} |z_+\rangle + \frac{2}{\sqrt{6}} |z_+\rangle + \frac{1}{\sqrt{3}} |z_-\rangle - \frac{1}{\sqrt{6}} |z_-\rangle \\ &= \frac{\sqrt{2}+1}{\sqrt{6}} |z_+\rangle + \frac{\sqrt{2}-1}{\sqrt{6}} |z_-\rangle = \begin{pmatrix} \frac{\sqrt{2}+1}{\sqrt{6}} \\ \frac{\sqrt{2}-1}{\sqrt{6}} \end{pmatrix} \end{aligned}$$

Ex: 3

$$|\psi\rangle = \frac{1}{2} |0_A\rangle |0_B\rangle + \frac{\sqrt{3}}{2} |0_A\rangle |1_B\rangle$$

$$\rho = |\psi\rangle\langle\psi| = \left(\frac{1}{2} |0_A\rangle |0_B\rangle + \frac{\sqrt{3}}{2} |0_A\rangle |1_B\rangle \right) \left(\frac{1}{2} \langle 0_A| \langle 0_B| + \frac{\sqrt{3}}{2} \langle 0_A| \langle 1_B| \right)$$

$$= \frac{1}{4} |0_A 0_B\rangle \langle 0_A 0_B| + \frac{\sqrt{3}}{4} |0_A 0_B\rangle \langle 0_A 1_B| + \frac{\sqrt{3}}{4} |0_A 1_B\rangle \langle 0_A 0_B| + \frac{3}{4} |0_A 1_B\rangle \langle 0_A 1_B|$$

$$\rho_A = \text{Tr}_B[\rho] = \frac{1}{4} |0_A\rangle \langle 0_A| \text{Tr}_B[|0\rangle\langle 0|] + \frac{\sqrt{3}}{4} |0_A\rangle \langle 0_A| \text{Tr}_B[|0\rangle\langle 1|] + \frac{\sqrt{3}}{4} |0_A\rangle \langle 0_A| \text{Tr}_B[|1\rangle\langle 0|] + \frac{3}{4} |0_A\rangle \langle 0_A| \text{Tr}_B[|1\rangle\langle 1|]$$

$$= \frac{1}{4} |0_A\rangle \langle 0_A| \overset{1}{\langle 0|0\rangle} + \frac{\sqrt{3}}{4} |0_A\rangle \langle 0_A| \overset{0}{\langle 0|1\rangle} + \frac{\sqrt{3}}{4} |0_A\rangle \langle 0_A| \overset{0}{\langle 1|0\rangle} + \frac{3}{4} |0_A\rangle \langle 0_A| \overset{1}{\langle 1|1\rangle}$$

$$= \frac{1}{4} |0_A\rangle \langle 0_A| + \frac{3}{4} |0_A\rangle \langle 0_A| = |0_A\rangle \langle 0_A| \Rightarrow \rho = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}_{|0_A\rangle, |1_A\rangle}$$

Ex. 4 $M = \begin{pmatrix} 1 & i & 0 \\ -i & 2 & -i \\ 0 & i & 1 \end{pmatrix}$

$$\det(M - \lambda I) = \begin{vmatrix} 1-\lambda & i & 0 \\ -i & 2-\lambda & -i \\ 0 & i & 1-\lambda \end{vmatrix} = (1-\lambda) \det \begin{pmatrix} 2-\lambda & -i \\ i & 1-\lambda \end{pmatrix} - i \det \begin{pmatrix} -i & -i \\ 0 & 1-\lambda \end{pmatrix} + 0 = (1-\lambda) [(2-\lambda)(1-\lambda) + i^2] - i [(-i)(1-\lambda) - 0]$$

$$= (1-\lambda)(2-2\lambda-\lambda+\lambda^2-1) + i^2-1^2\lambda = (1-\lambda)(\lambda^2-3\lambda+1) -1 + \lambda = \lambda^2 - 3\lambda - 1 - \lambda^2 + 3\lambda - 1 + \lambda = \lambda^2 - 3\lambda - 1 + \lambda^2 - 3\lambda - 1 + \lambda = 2\lambda^2 - 5\lambda - 2$$

$$\lambda^2 - 3\lambda - 2 = 0; \lambda(-\lambda^2 + 4\lambda - 3) = 0; -\lambda^2 + 4\lambda - 3 = 0 \quad \Delta: 16 - 4(3) = 4 \quad \frac{-4 \pm 2}{-2} < \lambda_2 = 3$$

$$\lambda_1 = 0$$

$$\lambda_2 = 3 \quad (0, 1, 3)$$

$$\lambda_3 = 1$$

Ex. 5 $\cdot |\psi(0)\rangle = |z\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $H = \begin{pmatrix} 2J & 0 \\ 0 & J \end{pmatrix}$ $\omega_0 = \frac{h}{J}; \omega_1 = \frac{h}{J}$ $e^{ix} = \cos(x) + i \sin(x)$

$$\cdot |\psi(t)\rangle = |z\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$|\psi(t)\rangle = U(t)|\psi(0)\rangle; \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{pmatrix} e^{-i\omega_0 t} & 0 \\ 0 & e^{-i\omega_1 t} \end{pmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}; \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} e^{-i\frac{2J}{\hbar}t} & 0 \\ 0 & e^{-i\frac{J}{\hbar}t} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix}; \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} e^{-i2Jt/\hbar} & 0 \\ 0 & e^{-iJt/\hbar} \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix};$$

$$\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} e^{-i2Jt/\hbar} \\ e^{-iJt/\hbar} \end{pmatrix}; \begin{cases} 1 = e^{-i2Jt/\hbar} & ; & 1 = \cos(-2Jt/\hbar) + i \sin(-2Jt/\hbar) \\ -1 = e^{-iJt/\hbar} & ; & -1 = \cos(-Jt/\hbar) + i \sin(-Jt/\hbar) \end{cases}$$

$$\cdot t = \frac{2}{J} \Rightarrow \begin{cases} 1 = \cos(-2) + i \sin(-2) & \ddot{} \\ -1 = \cos(-1) + i \sin(-1) & \ddot{} \end{cases}$$

$$\cdot t = \frac{\pi}{J} \Rightarrow \begin{cases} 1 = \cos(-2\pi) + i \sin(-2\pi) & \ddot{} \\ -1 = \cos(-\pi) + i \sin(-\pi) & \ddot{} \end{cases}$$

$$t = \frac{2\pi}{J} \Rightarrow \begin{cases} 1 = \cos(-4\pi) + i \sin(-4\pi) & \ddot{} \\ -1 = \cos(-2\pi) + i \sin(-2\pi) & \ddot{} \end{cases}$$

$$\cdot t = \frac{2}{J} \Rightarrow \begin{cases} 1 = \cos(-4) + i \sin(-4) & \ddot{} \\ -1 = \cos(-2) + i \sin(-2) & \ddot{} \end{cases}$$

$$\cdot t = 0 \Rightarrow \begin{cases} 1 = \cos(0) + i \sin(0) & \ddot{} \\ -1 = \cos(0) + i \sin(0) & \ddot{} \end{cases}$$

Ex. 6 $|\psi\rangle = \sqrt{\frac{2}{3}} |z_+\rangle + \frac{1}{\sqrt{3}} |z_-\rangle$

$$P = |\langle \psi | \psi \rangle| = \left| \left(\sqrt{\frac{2}{3}} \langle z_+ | + \frac{1}{\sqrt{3}} \langle z_- | \right) \left(\sqrt{\frac{2}{3}} |z_+\rangle + \frac{1}{\sqrt{3}} |z_-\rangle \right) \right| = \frac{2}{3} \langle z_+ | z_+ \rangle + \frac{\sqrt{2}}{3} \langle z_+ | z_- \rangle + \frac{\sqrt{2}}{3} \langle z_- | z_+ \rangle + \frac{1}{3} \langle z_- | z_- \rangle = \begin{pmatrix} 2/3 & \sqrt{2}/3 \\ \sqrt{2}/3 & 1/3 \end{pmatrix}$$

$$x = 2 \operatorname{Re}(P_{00}) = 2 \sqrt{2}/3$$

$$y = 2 \operatorname{Im}(P_{00}) = 2 \cdot 0 = 0 \quad \left(\frac{2\sqrt{2}}{3}; 0; \frac{1}{3} \right)$$

$$z = P_{00} - P_{11} = 2/3 - 1/3 = 1/3$$

Ex. 7 $\frac{1}{\sqrt{2}} (|1\downarrow\rangle - |1\uparrow\rangle)$

Ex: 8

$$a: \{ |0\rangle + |1\rangle; |0\rangle + |2\rangle; |2\rangle \}$$

CHECK NORMALIZATION

$$\langle a_1 | a_1 \rangle = (\langle 0| + \langle 1|) (|0\rangle + |1\rangle) = 1 + 1 = 2 \quad \checkmark$$

$$d: \left\{ \frac{|0\rangle - |1\rangle}{\sqrt{2}}; \frac{|0\rangle + |1\rangle}{\sqrt{2}}; |2\rangle \right\}$$

CHECK NORMALIZATION

$$\langle d_1 | d_1 \rangle = \frac{1}{2} (\langle 0| - \langle 1|) (|0\rangle - |1\rangle) = 1$$

$$\langle d_2 | d_2 \rangle = \frac{1}{2} (\langle 0| + \langle 1|) (|0\rangle + |1\rangle) = 1$$

$$\langle d_3 | d_3 \rangle = \langle 2|2\rangle = 1$$

CHECK ORTHONORMAL

$$\langle d_1 | d_2 \rangle = \frac{1}{2} (\langle 0| - \langle 1|) (|0\rangle + |1\rangle) = 0$$

$$\langle d_1 | d_3 \rangle = \frac{1}{\sqrt{2}} (\langle 0| - \langle 1|) |2\rangle = 0$$

$$\langle d_2 | d_3 \rangle = \frac{1}{\sqrt{2}} (\langle 0| + \langle 1|) |2\rangle = 0$$

$$\left\{ \frac{|0\rangle - |1\rangle}{\sqrt{2}}; \frac{|0\rangle + |1\rangle}{\sqrt{2}}; |2\rangle \right\}$$

$$b: \{ |0\rangle - |1\rangle; |0\rangle + |1\rangle; |2\rangle \}$$

CHECK NORMALIZATION

$$\langle b_1 | b_1 \rangle = (\langle 0| - \langle 1|) (|0\rangle - |1\rangle) = 0$$

$$\langle b_2 | b_2 \rangle = (\langle 0| + \langle 1|) (|0\rangle + |1\rangle) = 2 \quad \checkmark$$

$$c: \{ 2|0\rangle; |1\rangle; i|2\rangle \}$$

CHECK NORMALIZATION

$$\langle c_1 | c_1 \rangle = 4 \langle 0|0\rangle = 4 \quad \checkmark$$

$$e: \{ |0\rangle; |1\rangle + |2\rangle; |2\rangle \}$$

CHECK NORMALIZATION

$$\langle e_1 | e_1 \rangle = 1$$

$$\langle e_2 | e_2 \rangle = (\langle 1| + \langle 2|) (|1\rangle + |2\rangle) = 2 \quad \checkmark$$

Ex: 9

Pure $\Rightarrow \text{Tr}[\rho] = 1$ AND $\text{Tr}[\rho^2] = 1$ sum of eigenvalues of ρ^2

Ex: 10

$$O = 3|0\rangle\langle 0| - |1\rangle\langle 1| + (2+i)|0\rangle\langle 1| + (2-i)|1\rangle\langle 0| \quad |v\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

$$O = \begin{pmatrix} 3 & 2-i \\ 2+i & -1 \end{pmatrix} \quad |v\rangle = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} \quad \langle O O^\dagger \rangle = \langle O^2 \rangle - \langle O \rangle^2$$

$$\langle v | O | v \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 3 & 2-i \\ 2+i & -1 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 3 & 2-i \\ 2+i & -1 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \frac{1}{2} (1 \quad -1) \begin{pmatrix} 3-2+i \\ 2-i+1 \end{pmatrix} = \frac{1}{2} (1 \quad -1) \begin{pmatrix} 1+i \\ 3+i \end{pmatrix} = \frac{1}{2} (1+i-3-i) = -1$$

$$\begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 3 & 2-i \\ 2+i & -1 \end{pmatrix} \begin{pmatrix} 3 & 2-i \\ 2+i & -1 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & -1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 14 & 4-2i \\ 4+2i & 6 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix} = \frac{1}{2} (1 \quad -1) \begin{pmatrix} 14 & 4-2i \\ 4+2i & 6 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\begin{pmatrix} 9+4-i^2 & 6-3i-2+i \\ 6+3i-2-i & 4-i^2+1 \end{pmatrix}$$

$$= \frac{1}{2} (1 \quad -1) \begin{pmatrix} 14 & 4+2i \\ 4+2i & 6 \end{pmatrix} = \frac{1}{2} (1 \quad -1) \begin{pmatrix} 10+2i \\ -2+2i \end{pmatrix} = \frac{1}{2} (10+2i+2-2i) = \frac{1}{2} (12) = 6$$

$$\begin{bmatrix} \langle O \rangle = -1 \\ \langle O^2 \rangle = 6 \end{bmatrix} \quad \langle O O^\dagger \rangle = \langle O^2 \rangle - \langle O \rangle^2 = 6 - (-1)^2 = 5$$

SKIP IS LONG :)

$$\langle 4\sigma^2 \rangle = \langle \sigma^2 \rangle \cdot \langle \sigma \rangle^2$$

$$\langle \sigma \rangle = \langle \psi | \sigma | \psi \rangle = \frac{1}{\sqrt{2}} (\langle 0 | - \langle 1 |) \left(3|0x0\rangle - |1x1\rangle + (2+i)|0x1\rangle + (2-i)|1x0\rangle \right) \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle \right) = \left(\frac{1}{\sqrt{2}}\langle 0 | - \langle 1 | \right) \left(\frac{3}{\sqrt{2}}|0\rangle + \frac{3}{\sqrt{2}}|1\rangle - \frac{(2+i)}{\sqrt{2}}|0\rangle + \frac{2-i}{\sqrt{2}}|1\rangle \right)$$

$$= \frac{3}{2} - \frac{(2+i)}{2} - \frac{1}{2} - \frac{2-i}{2} = \frac{3-2-i-1-2+i}{2} = \frac{-2}{2} = -1$$

or TECHNICALLY IS -1

$$\langle \sigma \rangle^2 = 1^2 = 1$$

$$\langle \sigma^2 \rangle = \langle \psi | \sigma^2 | \psi \rangle =$$

$$\begin{aligned} \sigma^2 &= (3|0x0\rangle - |1x1\rangle + (2+i)|0x1\rangle + (2-i)|1x0\rangle) (3|0x0\rangle - |1x1\rangle + (2+i)|0x1\rangle + (2-i)|1x0\rangle) = \\ &= 9|0x0\rangle\langle 0x0| - 3|0x0\rangle\langle 1x1| + 3(2+i)|0x0\rangle\langle 0x1| + 3(2-i)|0x0\rangle\langle 1x0| \\ &\quad - 3|1x1\rangle\langle 0x0| + |1x1\rangle\langle 1x1| - (2+i)|1x1\rangle\langle 0x1| - (2-i)|1x1\rangle\langle 1x0| \\ &\quad + 3(2+i)|0x1\rangle\langle 0x0| - (2+i)|0x1\rangle\langle 1x1| + (2+i)(2+i)|0x1\rangle\langle 0x1| + (2+i)(2-i)|0x1\rangle\langle 1x0| \\ &\quad + 3(2-i)|1x0\rangle\langle 0x0| - (2-i)|1x0\rangle\langle 1x1| + (2-i)(2+i)|1x0\rangle\langle 0x1| + (2-i)(2-i)|1x0\rangle\langle 1x0| \\ &= 9|0x0\rangle + 3(2+i)|0x1\rangle + |1x1\rangle - (2-i)|1x0\rangle - (2+i)|0x1\rangle + (4-i^2)|0x0\rangle + 3(2-i)|1x0\rangle + (4-i^2)|1x1\rangle \\ &= (9+4+i)|0x0\rangle + (6-3i-2-i)|0x1\rangle + (6-3i-2-i)|1x0\rangle + (4+4+1)|1x1\rangle \\ &= 14|0x0\rangle + (4-i)|0x1\rangle + (4-2i)|1x0\rangle + 6|1x1\rangle \end{aligned}$$

$$\begin{aligned} \langle \psi | \sigma^2 | \psi \rangle &= \left(\frac{1}{\sqrt{2}}\langle 0 | - \langle 1 | \right) \left(14|0x0\rangle + (4-i)|0x1\rangle + (4-2i)|1x0\rangle + 6|1x1\rangle \right) \left(\frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle \right) = \\ &= \frac{1}{\sqrt{2}}\langle 0 | - \frac{1}{\sqrt{2}}\langle 1 | \left(\frac{14}{\sqrt{2}}|0\rangle - \frac{4-i}{\sqrt{2}}|0\rangle + \frac{(4-2i)}{\sqrt{2}}|1\rangle - \frac{6}{\sqrt{2}}|1\rangle \right) \\ &= \frac{14}{2} - \frac{4-i}{2} - \frac{4-2i}{2} + \frac{6}{2} = \frac{14-4+i-4+2i+6}{2} = \frac{12+3i}{2} = 6+1.5i \end{aligned}$$